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Question 1

Proof by Contradiction:
Assume that $N = pq$ is the product of two distinct primes and that N is a Carmichael number. Without loss of generality, assume $p > q$.

Theorem : Korselt's Criterion
A composite integer $N > 1$ is a **Carmichael number** if and only if:

1. N is square-free (i.e., N is not divisible by the square of any prime).
2. For every prime p dividing N, $p - 1$ divides $N - 1$.

We know that $p - 1 \mid pq - 1$ (since $p - 1$ divides $N - 1$, as per Korselt's criterion). This implies that there exists some integer $k \in \mathbb{N}$ such that:

$$pq - 1 = k(p - 1) \tag{1}$$

Additionally, we have:

$$pq - q = q(p - 1) \tag{2}$$

Now, subtract equation (2) from equation (1):

$$(p - 1)(k - q) = q - 1$$

This implies that $p - 1$ divides $q - 1$, i.e.,

$$p - 1 \mid q - 1$$

However, since we assumed that $p > q$, this is a contradiction because $p - 1$ cannot divide $q - 1$ under this assumption.
Thus, we reach a contradiction. Therefore, our assumption that N is a Carmichael number must be false. Hence, N cannot be a Carmichael number.

Korselt's Criterion

A composite integer $N > 1$ is a **Carmichael number** if and only if:

1. N is square-free (i.e., N is not divisible by the square of any prime).
2. For every prime p dividing N, $p - 1$ divides $N - 1$.

Below is a proof of Korselt's criterion.

Proof of Korselt's Criterion

Part 1: If N is a Carmichael number, then N satisfies Korselt's conditions

1. **N is square-free**:

   Suppose N is not square-free. Then there exists a prime p such that p^2 divides N. Let a be a primitive root modulo p^2.

   By the Chinese Remainder Theorem, there exists an integer a such that a is a primitive root modulo p^2 and $a \equiv 1 \pmod{N/p^2}$.

   Since N is a Carmichael number, $a^{N-1} \equiv 1 \pmod{N}$. This implies $a^{N-1} \equiv 1 \pmod{p^2}$.

   However, the multiplicative order of a modulo p^2 is $\phi(p^2) = p(p-1)$, which divides $N - 1$ only if p divides $N - 1$. But p divides N, so p cannot divide $N - 1$. This is a contradiction.

   Therefore, N must be square-free.

2. **For every prime p dividing N, $p - 1$ divides $N - 1$**:

   Let p be a prime divisor of N. Since N is square-free, p divides N exactly once. Let a be a primitive root modulo p. By the Chinese Remainder Theorem, there exists an integer a such that a is a primitive root modulo p and $a \equiv 1 \pmod{N/p}$. Since N is a Carmichael number, $a^{N-1} \equiv 1 \pmod{N}$. This implies $a^{N-1} \equiv 1 \pmod{p}$. The multiplicative order of a modulo p is $p - 1$, so $p - 1$ must divide $N - 1$.

Thus, if N is a Carmichael number, it satisfies Korselt's conditions.

Part 2: If N satisfies Korselt's conditions, then N is a Carmichael number

1. **N is square-free**:

   - By assumption, N is square-free, so $N = p_1 p_2 \cdots p_k$, where $p_1, p_2, \ldots, p_k$ are distinct primes.

2. **For every prime p dividing N, $p - 1$ divides $N - 1$**:

   Let a be an integer coprime to N. Then a is coprime to each p_i. By Fermat's Little Theorem, $a^{p_i - 1} \equiv 1 \pmod{p_i}$. Since $p_i - 1$ divides $N - 1$, we have $a^{N-1} \equiv 1 \pmod{p_i}$ for each i. By the Chinese Remainder Theorem, $a^{N-1} \equiv 1 \pmod{N}$.

Thus, N satisfies the definition of a Carmichael number.

1 Question 2

Proof:
Let G be a cyclic group of order n. and g_1 is generator Then we can write:

$$G = \langle g_1 \rangle = \{g_1^k \mid k \in \mathbb{Z}\}.$$

The element g_2 is also a generator, so g_2 can be written as some power of g_1:

$$g_2 = g_1^k.$$

and aslo we can written as

$$g_1^n = e \quad \text{and} \quad g_2^n = e.$$

For some integer k, where k is the discrete logarithm of g_2 with respect to g_1, we denote:

$$\log_{g_1} g_2 = k.$$

Substituting $g_2 = g_1^k$, we get:

$$(g_1^k)^n = g_1^{kn} = e.$$

The order of g_1 is n, so $g_1^m = e$ if and only if m is a multiple of n. From the equation:

$$g_1^{kn} = e,$$

we conclude:

$$kn \equiv 0 \pmod{n}.$$

This implies:

$$kn \text{ is a multiple of } n.$$

Let $d = \gcd(k, n)$. We want to show that $d = 1$.
Since d divides both k and n, we can write:

$$k = d \cdot k', \quad n = d \cdot n',$$

where $\gcd(k', n') = 1$.
Substituting into $g_2 = g_1^k$, we get:

$$g_2 = g_1^{d \cdot k'}.$$

The order of g_2 is n, so:

$$(g_1^{d \cdot k'})^n = g_1^{d \cdot k' \cdot n} = e.$$

Since the order of g_1 is n, we have:

$$d \cdot k' \cdot n \equiv 0 \pmod{n}.$$

Simplifying, this implies:

$$d \cdot k' \equiv 0 \pmod 1.$$

Since $\gcd(k', n') = 1$, the only possibility is $d = 1$.
Thus, we conclude:

$$\gcd(\log_{g_1} g_2, n) = 1.$$

Question 3

The group $(\mathbb{Z}/p^n\mathbb{Z})^\times$ consists of integers modulo p^n that are coprime to p^n. The order (size) of this group is given by Euler's totient function:

$$\varphi(p^n) = p^n - p^{n-1} = p^{n-1}(p-1),$$

which is the number of integers in $\{1, 2, \ldots, p^n - 1\}$ that are relatively prime to p^n.
Counting Elements of Even Order
The order of any element in $(\mathbb{Z}/p^n\mathbb{Z})^\times$ must divide the group's order $\varphi(p^n) = p^{n-1}(p-1)$. We need to count elements whose order is even.

Case 1: $p = 2$ (Power of 2 Modulo)

If $p = 2$, then: using Euler's totient function for 2^n is given by:

$$\varphi(2^n) = 2^n - 2^{n-1} = 2^{n-1}$$

which is always a power of 2, meaning all elements have odd order. Thus, there are zero elements of even order.

Case 2: p is an Odd Prime

For an odd prime p, the group's order is:

$$\varphi(p^n) = p^{n-1}(p-1).$$

Since $p - 1$ is even, this means that half of the elements in the group must have even order (because any finite group of even order has exactly half of its elements with even order).
Thus, the number of elements with even order is:

$$\frac{\varphi(p^n)}{2} = \frac{p^{n-1}(p-1)}{2},$$

which is an integer because $p - 1$ is even.

Final Answer

- If $p = 2$, then 0 elements have even order.
- If p is an odd prime, then the number of elements with even order is:

$$\frac{p^{n-1}(p-1)}{2}.$$